

Urban Air Quality Early Warning System

Mid Semester Project Progress Report

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Executive Summary

This report records the mid-semester status of a two-pollutant forecasting prototype for Delhi. The application now acquires and caches modeled CAMS PM2.5 and PM10, reconstructs a separate R.K. Puram station series from two public archives after an hourly overlap audit, and tests chronological forecasts against simple baselines. The strongest current result is station PM10: the selected Random Forest achieved 27.61 ug/m3 mean absolute error (MAE), compared with 37.41 ug/m3 for persistence on 133 scored dates. Persistence remained the selected method for the other three source-pollutant combinations. These are historical development results, not a verified claim about future live accuracy.

1 Project Objective and Current Status

The project asks whether weather, pollution history, and known calendar events improve daily PM2.5 and PM10 forecasts beyond carrying forward the previous day's concentration. The prototype predicts concentration in micrograms per cubic metre (ug/m3), not an AQI score. It compares predictions with the Indian 24-hour standards of 60 ug/m3 for PM2.5 and 100 ug/m3 for PM10 [5].

Workstream	Mid semester status
Historical acquisition and SQLite caching	Implemented and populated locally
Station archive overlap audit	Passed preset gate for both pollutants
Shared modeling and chronological backtests	Implemented and tested
Seven-day Streamlit outlook	Implemented with stale-station CAMS fallback
Prospective live accuracy record	Not yet mature enough to report

2 Data Provenance and Coverage

CAMS concentrations are atmospheric model output obtained through Open-Meteo [1]. They are never presented as CPCB station measurements. R.K. Puram observations come from the CPCB CAAQM network as compiled by XKDR [2] and archived by OpenAQ [3]. Historical and forecast weather come from Open-Meteo. The data sources retain separate target columns throughout the application.

Target	Five year calendar span	Valid days
CAMS modeled PM2.5	2021-10-09 to 2026-10-08	1,526 / 1,826
CAMS modeled PM10	2021-10-09 to 2026-10-08	1,526 / 1,826
R.K. Puram station PM2.5	2021-10-06 to 2026-10-05	1,610 / 1,826
R.K. Puram station PM10	2021-10-06 to 2026-10-05	1,594 / 1,826

The five-year label refers to a calendar span. It does not mean five complete years of daily measurements. The CAMS global archive starts in August 2022 [1]. The station series also has missing or incomplete days. The application does not impute particulate targets.

3 Source Quality Controls

The two station archives were compared on 6,167 matched local hourly readings between 19 February and 1 September 2025. The preset merge gate required at least 95% of matched readings for each pollutant to differ by no more than 1 ug/m3 and an absolute median signed difference no greater than 1 ug/m3. Both pollutants passed.

Pollutant	Matched hours	Within 1 ug/m3	Median difference
PM2.5	3,099	98.9%	0.0 ug/m3
PM10	3,068	97.7%	0.0 ug/m3

Unmatched hours remain in both archives and are recorded in the saved audit. OpenAQ's raw unit text contains replacement characters, so exact byte-level unit-string equivalence could not be verified. Each station pollutant needs at least 18 distinct valid local hours to form a daily mean. CAMS daily values require 24 numeric hourly values for both pollutants and required weather.

4 Modeling and Validation Design

The shared feature set contains 21 variables: weather, date-aware PM lags of 1, 2, 3 and 7 calendar days, a complete prior seven-day mean, seasonal terms, weekend and Indian holiday indicators, and Diwali proximity. Wind direction is encoded with sine and cosine. Calendar-day joins prevent a missing day from becoming a false previous-day value. Optional satellite fire detections may be cached but are not included in this five-year comparison.

For each training window, predictor imputation, Isolation Forest anomaly detection, scaling, Ridge, and Random Forest are fitted using training dates only. The anomaly-filtered model falls back to all usable training rows if fewer than 10 normal rows remain. Persistence, a trailing seven-day mean, and training-only day-of-year climatology serve as simpler comparators.

Earlier seasonal windows select the production method. The latest 180 calendar days are then evaluated chronologically on identical scored dates. Each one-day-ahead test prediction uses the true previous calendar day's PM, which would be known in next-day operation. A separate seven-day outlook recursively feeds earlier predictions into later lags. The final test uses Open-Meteo's stitched historical-forecast weather [4]. That archive is not a fixed weather forecast issued one day before every target date.

5 Preliminary Results

Target	Scored	Selected	MAE	Persistence
CAMS modeled PM2.5	180 / 180	Persistence	16.62	16.62
CAMS modeled PM10	180 / 180	Persistence	86.99	86.99
R.K. Puram station PM2.5	131 / 180	Persistence	12.12	12.12
R.K. Puram station PM10	133 / 180	Random Forest	27.61	37.41

The station PM10 Random Forest reduced final-test MAE by 26.2% relative to persistence. Its R2 was 0.732; exceedance recall was 92.3%, precision 88.4%, and false-positive rate 26.2% at a 68.4% exceedance base rate. CAMS PM2.5 and PM10 retained persistence because the more complex candidates did not pass the earlier seasonal promotion rule.

Station PM2.5 illustrates why the selection rule matters. Random Forest had a slightly lower MAE than persistence on the final period (11.52 versus 12.12), but it improved only 7 of 10 earlier seasonal windows. The preset rule required at least 8. The final-period result did not change the selected method. MAE values should be compared within a pollutant and source, not across PM2.5 and PM10 concentration scales.

6 Application Deliverable and Verification

The Streamlit application presents coverage, source-specific backtests, recursive seven-day forecasts, effective lead times, and threshold warnings. If the station observation is more than five days old or its model cannot issue a forecast, the app displays a CAMS outlook. Local SQLite caches support repeat launches, while results.json preserves an inspectable snapshot without distributing raw caches or API keys.

As of this report, 33 focused unit tests passed. Streamlit AppTest passed for all four source-pollutant combinations. The interface was checked in desktop and mobile browser widths. The repository is available at <https://github.com/yashvardhansingroha/N2.5>.

7 Limits and Next Phase

- Collect immutable forecasts before target dates and pair them later with observations. Only this can support a prospective live-accuracy claim.
- Evaluate archived weather from a fixed forecast issue time. Stitched near-term weather can make a historical test optimistic.
- Continue assessing station gaps, source-unit metadata, and changes in station coverage. Preserve separate modeled and observed targets.
- Assess satellite fire signals only after a sufficiently complete historical archive supports leakage-safe seasonal backtests.

The 180-day period is excluded from fitting and programmatic method selection, but its metrics were inspected during development. The results are therefore historical development evidence, not a pristine one-time blind trial or an official health advisory.

References

- [1] Open-Meteo. Air Quality API documentation. <https://open-meteo.com/en/docs/air-quality-api>
- [2] XKDR Forum. India Air Quality Database and API. <https://airquality.xkdr.org/>
- [3] OpenAQ. Open Data on AWS archive documentation. <https://docs.openaq.org/aws/about>
- [4] Open-Meteo. Historical Forecast API documentation. <https://open-meteo.com/en/docs/historical-forecast-api>
- [5] Central Pollution Control Board. National Ambient Air Quality Standards, 24-hour values. https://cpcb.gov.in/upload/NAAQS_2019.pdf
- [6] Project results snapshot, 09 October 2026, 13:51 IST. Local results.json in the project repository.